

AI Student Support System

TNSDC Virtual Internship Program

Track / Domain: Agentic AI

Core Methodology: Tool Calling + Vector RAG + Memory

1. Project Overview

Project Title: AI Student Support System

Problem Statement: Collegiate environments suffer from fragmented rulebooks, delay-prone administrative desks, and keyword-based search portals that fail when students query using colloquial terms (e.g., *bunk*, *od*, *gatepass*). Moreover, general LLMs often hallucinate on quantitative academic thresholds.

Brief Description: An autonomous agentic application combining an Alias Mapping layer, Vector-based RAG (Cosine Similarity), and a deterministic Python arithmetic tool to address campus queries and attendance eligibility verification without factual hallucinations.

2. Objectives & Proposed Solution

Project Objectives:

- Build an autonomous routing system capable of differentiating between policy searches, calculations, and general conversation.
- Eliminate hallucinations by grounding responses in verified college regulations across 7 campus domains.
- Enable deterministic verification for examination attendance clearance (75% standard, 65% condonation).

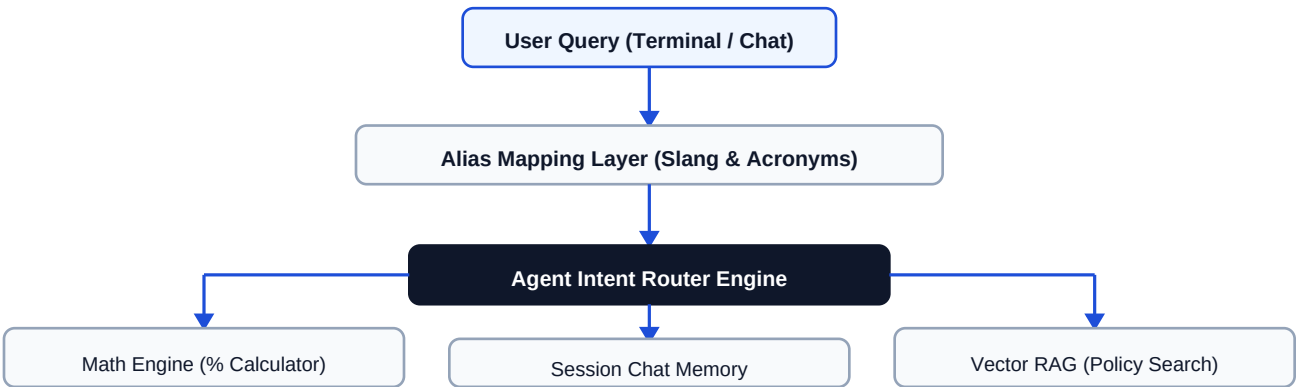
How the Agentic AI Solution Works: User inputs are intercepted by an Alias Resolution layer that normalizes colloquial student terms into system-indexed concepts. The Agent Intent Router then decides whether to invoke the Quantitative Math Tool (for numbers and attendance calculations) or the Semantic Vector RAG Engine (for regulations and guidelines).

Key Features: Natural language slang normalization, exact attendance computation, grounded multi-rule retrieval for broad requests, and session memory context tracking.

3. Implementation & Results

Technologies / Tools Used: Python 3.10+, LangChain Core, Cosine Vector Similarity Engine, Regular Expression (Regex) Parsers, Math & Counter Libraries.

Working Process & Architecture:



Screenshots / Test Outputs (Results Achieved):

TC #	Student Input	Engine Triggered	System Output	Status
TC-01	policy	Broad Overview	Compiles multi-category handbook summary across Academics, Grooming, Hostel, Leave.	PASS
TC-02	I attended 45 out of 60 classes	Math Tool + Policy	75.00% calculated. Output: ELIGIBLE. Accompanied by official regulation clause.	PASS
TC-03	I attended 38 out of 55 classes	Math Tool	69.09% calculated. Flags CONDONATION REQUIRED with Principal medical clearance.	PASS
TC-04	What are the rules for od?	Alias Map + RAG	Resolves 'od' -> 'on_duty'. Retrieves 14-day symposium/paper presentation rule.	PASS
TC-05	Can I keep a beard or long hair?	Alias Map + RAG	Maps terms -> 'grooming'. Returns male student hair trimming & shave mandates.	PASS
TC-06	Can I get an outing pass tonight?	Alias Map + RAG	Maps 'outing' -> 'hostel_leave'. Returns 24-hr warden sanction prerequisite.	PASS

4. Conclusion & Future Scope

Project Conclusion: The AI Student Support System successfully fulfills the IBM Agentic AI internship requirements. Combining deterministic tools with alias-enhanced vector search prevents hallucinations and provides instantaneous, accurate academic guidance.

Challenges Faced: Resolving student slang variations without bloating vector databases, and ensuring two-letter abbreviations (e.g., *od*, *ml*, *tc*) were not dropped by standard text splitters.

Future Enhancements: Developing a browser-based Streamlit UI, implementing automated PDF notice parsing, and adding regional multilingual speech support for campus kiosks.

5. References

- LangChain Documentation: Building Tool-Calling & ReAct Agents.
- IBM SkillsBuild: Agentic AI & Generative AI Foundations Curriculum.
- TNSDC Virtual Internship Program: Applied Agentic Systems Guidelines.